SYMBIOSIS INSTITUTE OF TECHNOLOGY, HYDERABAD
Constituent of Symbiosis International (Deemed University), Pune
(Established under Section 3 of the UGC Act of 1956 vide notification number F.9-12/2001-U.3 of the Government of India)
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USE-CASE REPORT

Breast Cancer Diagnosis Using Supervised Machine Learning

Submitted by

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24070721014 - CSE (V Sem)

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Abstract

Breast cancer diagnosis is an important application of supervised machine learning, where classification algorithms can be evaluated for their ability to distinguish malignant from benign tumor samples. This case study investigates the performance of five supervised learning algorithms using the Wisconsin Diagnostic Breast Cancer (WDBC) dataset available through the UCI Machine Learning Repository. The dataset contains 569 observations and 30 numerical features extracted from digitized images of cell nuclei.

The study follows a systematic experimental workflow covering dataset examination, exploratory data analysis, missing-value assessment, outlier detection, feature preprocessing, baseline modeling, advanced ensemble learning, hyperparameter optimization, and performance interpretation. The dataset was divided into 70% training and 30% testing subsets using stratified sampling. Logistic Regression, Decision Tree, and K-Nearest Neighbors were implemented as baseline classifiers, while Random Forest and Gradient Boosting were evaluated as advanced models. Five-fold stratified cross-validation was used to assess training performance, and GridSearchCV was applied to optimize the Random Forest classifier.

The results demonstrate that the classification problem can be modeled effectively using the available nuclear measurements. Logistic Regression achieved 97.66% testing accuracy, 98.39% precision, 95.31% malignant-class recall, an F1 score of 0.9683, and an ROC AUC of 0.9969. The optimized Random Forest achieved the same testing accuracy of 97.66%, an F1 score of 0.9677, and a slightly higher ROC AUC of 0.9980. However, Logistic Regression demonstrated better average cross-validation F1 performance and identified one additional malignant sample in the held-out test set.

The findings indicate that model complexity does not necessarily translate into superior classification performance. For this particular dataset and experimental configuration, Logistic Regression provides a competitive and relatively interpretable solution. The study also discusses the importance of false-negative errors, performance uncertainty, model generalization, and the limitations of using historical benchmark data for healthcare applications.

1. Problem and Objective

Machine learning has become an important area of computational research because it enables the identification of relationships within complex datasets. In supervised learning, an algorithm learns a mapping between input variables and known target labels using previously observed examples. The trained model can subsequently generate predictions for unseen observations. Supervised learning problems are generally categorized into regression and classification. Regression predicts continuous numerical values, whereas classification assigns observations to predefined categories. Medical diagnosis provides a useful setting for studying classification algorithms. Clinical measurements frequently contain multiple related variables, and the relationship between these variables and diagnostic outcomes may not be immediately apparent through simple observation.

Breast cancer diagnosis is one example in which numerical measurements can be used to investigate differences between benign and malignant tumor samples. In the Wisconsin Diagnostic Breast Cancer dataset, these measurements describe characteristics of cell nuclei observed in digitized images of fine-needle aspirate samples.

The objective of the present work is to investigate how effectively conventional supervised learning algorithms can classify these observations. The study is designed as an academic machine learning experiment. Although

the dataset originates from a medical context, the resulting models are not intended to replace medical professionals or provide clinical diagnoses.

2. Dataset and Exploratory Analysis

The Wisconsin Diagnostic Breast Cancer dataset was obtained through scikit-learn's local copy of the UCI repository dataset [1, 2]. All 30 features are continuous numerical measurements, including radius, texture, perimeter, area, concavity and concave points. The dataset labels were recoded from the library convention so that 1 denotes malignant and 0 benign; this makes positive-class recall unambiguous.

Dataset characteristic	Observed value
Total samples	569
Predictor variables	30 numeric
Benign / malignant	357 / 212
Missing values	0
Duplicate predictor rows	0
Training / test examples	398 / 171

No values required imputation and there were no duplicate predictor rows. The standard 1.5 × IQR rule flagged 608 cells across all feature columns, not 608 distinct patients. These observations were retained because extreme measurements can reflect the very condition under study. Histograms and correlations were used to inspect skew and associations; correlation is not evidence of causation.

The feature groups are also important for judging whether a classifier is learning distinct signals or several descriptions of the same underlying structure. Mean radius, mean perimeter and mean area are closely related geometric quantities, while measures such as concavity and concave points characterize shape irregularity. A high absolute correlation with the target is useful for exploration, but it does not establish that a predictor is independently important once the remaining features are considered. For this reason, the EDA is treated as descriptive rather than as a mechanism for selecting features from the full dataset before splitting. All 30 numerical measurements were retained for the experiments.

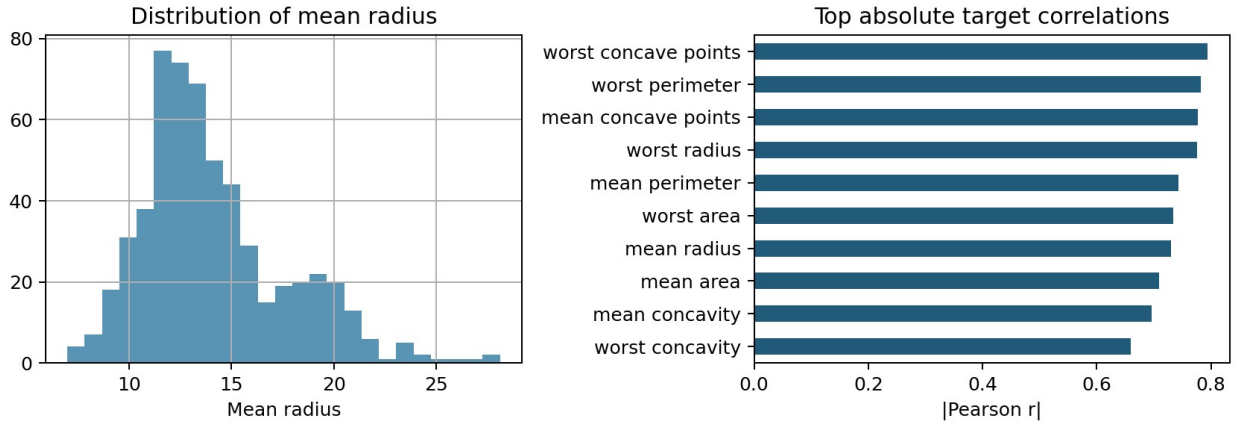


Figure 1 Mean radius distribution and strongest absolute correlations with the target

3. Methodology

A stratified random split (seed 42) reserved 30% of records for testing. Training-only pipelines applied median imputation to all classifiers and standardized the input features for Logistic Regression and KNN. Neither scaling nor imputation was fitted using test data. Five-fold StratifiedKFold cross-validation on the training

partition provided mean F1 scores. The held-out test partition was then used to report the common set of comparison metrics.

The order of operations was chosen to avoid contamination of validation results. The hold-out split was created before fitting any preprocessing objects. Median values and scaling parameters were estimated only within the training partition or training fold to which they belonged. The test observations therefore did not contribute to estimated feature means, standard deviations, or fitted classifier coefficients. The same five-fold stratification and F1 scoring convention were used when comparing the training stability of the candidate algorithms. No categorical encoding was necessary because all predictors were numeric.

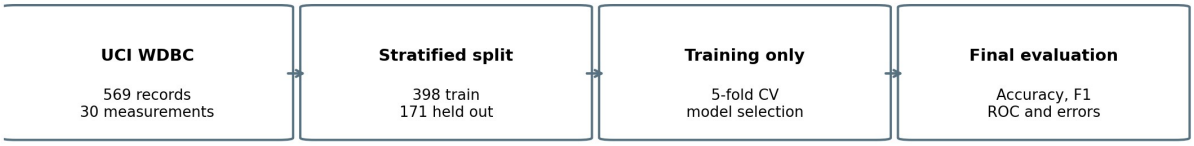


Figure 2 Data flow and separation of model training from final testing

Logistic Regression is a linear probabilistic classifier; Decision Tree learns sequential threshold rules; and KNN predicts from nearby training samples. Random Forest averages many randomized trees, while Gradient Boosting builds trees sequentially to correct earlier errors. The same input features, target coding and test records were used for all five models. Balanced class weights were applied to Logistic Regression, Decision Tree and Random Forest; SMOTE was not necessary because the faculty requirement permits class weights as an alternative.

The five algorithms provide different modeling assumptions. Logistic Regression estimates a linear boundary in the standardized feature space, whereas KNN classifies an observation using local neighborhood structure. The constrained Decision Tree learns a comparatively small set of threshold rules. Random Forest reduces some of the instability of individual trees by aggregating an ensemble, and Gradient Boosting constructs trees sequentially to reduce loss. Comparing these methods is useful because it tests whether the more flexible nonlinear approaches produce a meaningful advantage over a simpler linear model on the same data.

4. Baseline and Ensemble Results

Table 2 gives metrics for the malignant class (positive label = 1). The cross-validation F1 column is from the training data only. Accuracy, precision, recall, F1 and ROC AUC are from the 171 held-out test samples. This distinction matters because the cross-validation score is the more appropriate internal measure of how consistently the models trained.

Classifier	Accuracy	Precision	Recall	F1	AUC	CV F1
Logistic Regression	0.977	0.984	0.953	0.968	0.997	0.957
Decision Tree	0.912	0.902	0.859	0.880	0.898	0.900
KNN	0.965	1.000	0.906	0.951	0.988	0.943
Random Forest	0.971	1.000	0.922	0.959	0.997	0.922
Gradient Boosting	0.965	1.000	0.906	0.951	0.997	0.912

Logistic Regression tied the highest test accuracy (97.7%) and achieved the highest mean training-fold F1 (0.957). Its malignant recall of 95.3% exceeded that of the untuned forest (92.2%) and KNN (90.6%). Decision Tree performed least well, consistent with the high variance of a single shallow tree. Gradient Boosting had an excellent AUC, but its recall and F1 at the default decision threshold were lower than Logistic Regression. An AUC comparison does not by itself establish which decision threshold should be used.

The metrics should be read together. Precision measures how frequently predicted malignant cases are truly malignant; recall measures how many actual malignant cases were detected. A classifier can report perfect precision by making fewer positive predictions, yet miss more malignant observations. This trade-off is visible in the KNN and ensemble results. Mean training-fold F1 provides an additional criterion because it summarizes performance over multiple validation subsets, although it remains an estimate from this single dataset rather than an independent external validation result.

5. Hyperparameter Search and Advanced Evaluation

A class-weighted Random Forest was tuned using GridSearchCV. The grid evaluated 100 or 200 trees, a maximum depth of 6, 12 or no limit, and a minimum leaf size of 1 or 3. This produced 12 configurations evaluated in five stratified training folds. The chosen parameters were 100 trees, unrestricted depth and a minimum leaf size of 3. The best training-fold mean F1 was 0.927.

The grid included two choices for the number of trees (100 and 200), three maximum-depth settings (6, 12 and unrestricted), and two leaf-size settings (1 and 3). This amounts to 12 configurations and 60 cross-validation training-and-scoring evaluations. The selection criterion was F1 for the malignant class, reflecting the need to consider both missed malignancies and false alerts. The selected settings are best only within this finite search grid; a wider search or a different cross-validation partition could favor other parameters. Importantly, no hyperparameters were selected from the held-out testing labels.

The tuned forest improved its test F1 from 0.959 to 0.968 and increased malignant recall from 92.2% to 93.8%. On the held-out records it correctly identified 60 malignant cases, missed 4 malignant cases, correctly identified 107 benign cases and generated no false malignant alerts. Its ROC AUC was 0.998. These are the results of one partition, not a confidence guarantee or an estimate of performance at other hospitals.

The optimized forest is an instructive result because it improves upon the original forest on this test partition but does not dominate every competing method. Its test precision is 1.000, which corresponds to zero false malignant alerts among the 171 reserved observations. Its recall of 0.9375 corresponds to four missed malignant samples out of 64. In contrast, the Logistic Regression model identified 61 of those 64 malignant samples, with one additional benign sample incorrectly flagged. Equal accuracy thus conceals different error distributions; for this case study, the underlying counts are more informative than reporting accuracy alone.

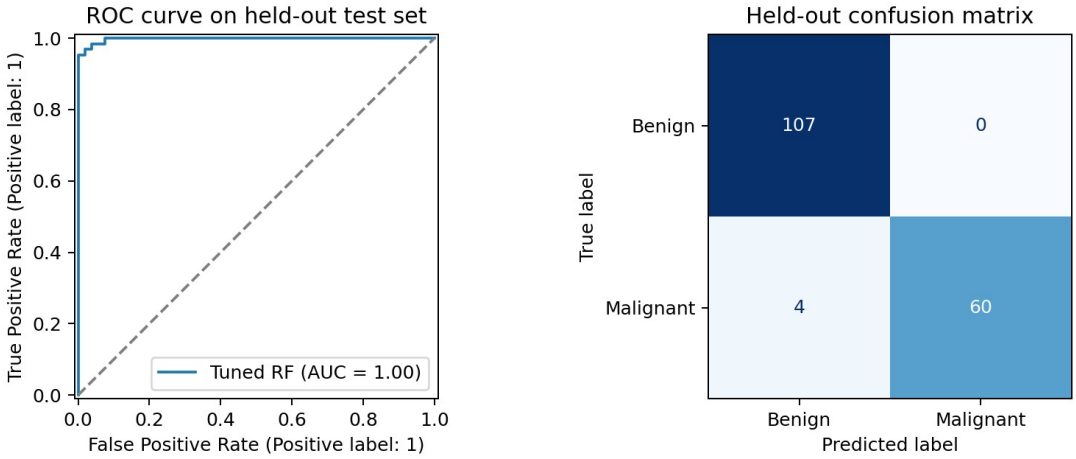
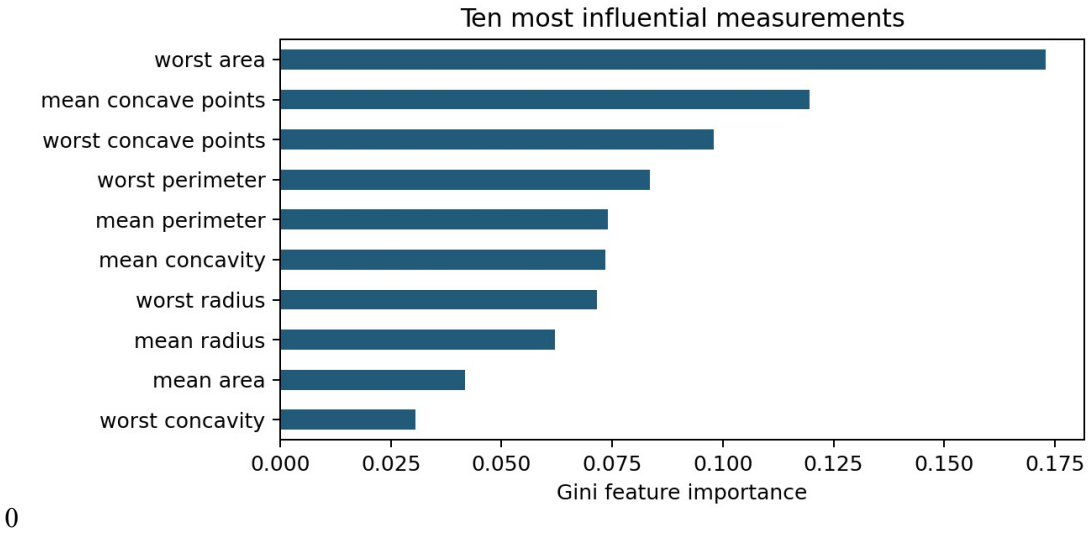


Figure 3 Tuned Random Forest ROC curve and confusion matrix on the held-out set

6. Interpretation and Model Selection

The tuned forest showed competitive performance, but it was not the best choice under the stated priorities. Logistic Regression yielded mean cross-validated F1 0.957, compared with 0.927 for the tuned forest, and test-set malignant recall 95.3%, compared with 93.8%. It is also comparatively simple to inspect. We therefore select Logistic Regression for the final academic demonstration, rather than choosing the most complicated algorithm or the one with the highest single AUC.

The forest feature-importance plot remains useful for understanding the advanced model. Radius, perimeter and concave-point measurements are related, so their scores may reflect overlapping information. Gini importance ranks influence on tree splits; it does not show that a feature causes disease or that the model would remain accurate after distribution changes.



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Figure 4 Top ten impurity-based predictors from the optimized Random Forest

7. Limitations, Practical Value and Future Work

The dataset is small and historical, with one source of imaging-derived measurements. No independent hospital, scanner or population was available for external validation. A single train/test partition can produce optimistic or pessimistic metrics by chance. The model is also not tested for calibration, demographic fairness or robustness to acquisition changes. Although class weights respond to the observed class proportions, they do not remove the need for threshold or cost analysis in a real decision process.

A sensible extension would begin with repeated nested cross-validation, external data and confidence intervals for sensitivity. A calibrated probability model could then be assessed at several thresholds with a clinician-defined false-negative cost. Other useful checks are permutation importance, subgroup performance, missing-input tests and versioned monitoring. Only after independent clinical evaluation should a user interface be considered beyond an educational demonstration.

8. Statistical Interpretation of the Test Results

The 171-record hold-out sample supports a comparison of model predictions, but it is not large enough to establish small differences as definitive. For the selected Logistic Regression model, 167 of 171 diagnoses were correctly classified. Of the 64 malignant samples, 61 were correctly predicted; the other three were false negatives. The corresponding sensitivity is 61/64 = 95.31%, while specificity is 106/107 = 99.07%. These two rates answer different questions: sensitivity concerns missed malignancies, whereas specificity concerns false alarms among benign samples. The positive predictive value is 61/62 = 98.39%.

A useful check on the apparent precision of these numbers is the Wilson binomial interval. Using a nominal 95% interval, accuracy is approximately 94.1% to 99.1% and malignant recall is approximately 87.1% to 98.4%. The interval for the tuned Random Forest malignant recall (60/64) is about 85.0% to 97.5%. The intervals overlap substantially. They summarize sampling uncertainty under a simplified binomial assumption and do not account for changes across institutions or for the process of comparing several candidate models. Consequently, the one-case recall advantage of Logistic Regression should be reported as an observed result, not as proof of general superiority.

The optimized Random Forest had a slightly larger ROC AUC (0.9980 versus 0.9969 for Logistic Regression). AUC summarizes how well scores rank positive examples across many thresholds, while recall and precision are calculated after applying a particular decision threshold. A model can therefore have a higher AUC and still detect fewer malignant cases at its default threshold. A clinical evaluation would need a specified action threshold and a numerical cost or utility associated with each error type. In this assignment, the default decision threshold is retained so that the algorithms can be compared under a consistent, transparent setup.

The F1 score is the harmonic mean of precision and recall: F1 = 2 × precision × recall / (precision + recall). It penalizes models that achieve one of those measures at the expense of the other, but it does not explicitly encode the higher cost that a missed malignant diagnosis might carry. Balanced class weights in Logistic Regression and Random Forest change the training objective; they do not guarantee balanced clinical consequences. This is why class-level confusion matrices are presented alongside summary metrics. The distinction matters even on this dataset, where both leading models have the same rounded accuracy but make different types of errors.

For reference, the Logistic Regression test confusion matrix can be written as 106 true benign, 1 false malignant alert, 3 missed malignant samples, and 61 correctly detected malignant samples. These counts yield sensitivity 61/64 = 95.31%, specificity 106/107 = 99.07%, and positive predictive value 61/62 = 98.39%. The tuned forest instead recorded 107 true benign, 0 false alarms, 4 missed malignant samples, and 60 true malignant detections. The difference is only one observation in each error category. Wilson intervals help convey the sampling uncertainty, but formal evidence of model superiority would require a more substantial design with independent evaluation data.

9. Validation Design and Threats to Generalization

The main protection against information leakage in this experiment is the pipeline construction. Imputation and scaling are fitted separately inside each training fold, so the validation fold does not influence the preprocessing parameters. The separate 30% test partition is never used by GridSearchCV. This design is considerably more reliable than scaling the full dataset before the split, which would inadvertently use statistics from observations reserved for evaluation. Stratification also helps preserve the approximately 37% malignant proportion across partitions, reducing variation caused by accidental class imbalances.

Nevertheless, there is an important limitation to the model-comparison procedure. The five candidate algorithms are described using their held-out test scores, and the final explanation considers both cross-validation and test-set recall. Repeatedly inspecting a test set while making modelling decisions can introduce selection bias, even if the algorithms are not directly trained on the test examples. The present results should therefore be understood as a reproducible classroom comparison. A stronger study would reserve an additional external test set, decide the selection criterion in advance and apply nested cross-validation: the inner loop would tune hyperparameters, while the outer loop would estimate generalization across independent held-out folds.

Data provenance and intended use are equally important. WDBC records represent digitized fine-needle aspirate images summarized into engineered nuclear features, not raw patient symptoms, mammography scans or general screening records. A model trained on these attributes cannot directly accept ordinary clinical history or unprocessed images. Its measured performance also says nothing about how well it works with different microscopes, staining protocols, image-processing pipelines, demographic populations or disease prevalence.

Any proposal to transfer these results into healthcare practice would first need representative independent datasets and professional governance.

Future validation should include a pre-registered primary metric, confidence intervals obtained through patient-level resampling where appropriate, probability calibration, sensitivity at clinically meaningful thresholds and error review by qualified experts. If multiple records could represent one individual in another dataset, grouping by patient would be essential to prevent leakage between training and testing. Model explanation should be examined with correlated predictors in mind: radius, perimeter and area describe related aspects of nucleus size, and a high tree-importance value should not be read as independent biological causation. Calibration plots, permutation importance and subgroup analyses would improve the evidence beyond the present course requirements.

The existing comparison is therefore best described as a transparent classroom benchmark. It demonstrates correct model construction, held-out evaluation, reproducible tuning and careful error interpretation, but it cannot estimate the consequences of using a prediction in a hospital. A subsequent investigation would need to specify the intended population, feature acquisition procedure, primary endpoint and acceptable error costs before model development. It would also need independent data collected under that intended workflow. Such a study may find that calibration or sensitivity at a chosen threshold matters more than a small difference in default-threshold accuracy.

10. Reproducibility and Error Discussion

To reproduce the experiment, open the accompanying .ipynb file in JupyterLab or Google Colab and run all cells in order. The notebook imports numpy, pandas, matplotlib, scikit-learn and IPython. It loads the dataset directly from scikit-learn, so no external CSV or login is needed. Random state 42 is used for the train/test split, cross-validation shuffling and the applicable estimators. All transformations are fitted inside pipelines. Grid search compares 12 Random Forest configurations using five training folds; the test partition remains separate from tuning. The notebook supplied with this report has already been executed and preserves its outputs.

For the selected Logistic Regression classifier, the held-out confusion matrix contained 106 correctly identified benign samples, one benign sample classified as malignant, 61 correctly identified malignant samples and three malignant cases classified as benign. In this data split the selected model therefore produced one false positive and three false negatives. The tuned Random Forest instead produced no false positives and four false negatives. This contrast explains the final choice: when missed malignancies are especially concerning, the extra correctly identified malignant sample can matter more than the additional benign false alarm. This is only a qualitative task preference; it is not a validated clinical cost function.

The reported accuracy of 97.66% should be read together with the class-level errors. A small count of missed cases can coexist with high accuracy because the majority class is benign. Model ranking is also sensitive to the chosen sample split. For a future study, the test examples should be set aside before any model comparison, then candidate selection should rely on nested cross-validation and an independent external dataset should be used for final claims. No external validation was available here. These constraints are part of the interpretation of the results, not evidence that the models are unreliable in every context.

11. Conclusion

This case study investigated the application of supervised machine learning algorithms to the classification of malignant and benign tumor samples using the Wisconsin Diagnostic Breast Cancer dataset.

The experimental procedure included exploratory data analysis, missing-value assessment, outlier detection, feature preprocessing, stratified data partitioning, baseline modeling, ensemble learning, hyperparameter tuning, cross-validation, and performance interpretation.

Five classification algorithms were evaluated under a common experimental configuration.

Logistic Regression achieved an accuracy of 97.66%, precision of 98.39%, malignant recall of 95.31%, F1 score of 0.9683, and ROC AUC of 0.9969.

The optimized Random Forest achieved the same testing accuracy and a slightly higher ROC AUC of 0.9980. However, its malignant recall and mean cross-validation F1 score were lower than those of Logistic Regression.

The comparison demonstrates that more complex algorithms do not necessarily produce better results for every dataset.

The study also highlights the limitations of relying exclusively on accuracy. Confusion matrices, recall, precision, and uncertainty analysis provide a more informative assessment of classification behavior.

Based on the observed results, Logistic Regression was selected as the preferred classifier for this educational experiment because it combines strong predictive performance, comparatively straightforward interpretation, and higher malignant recall within the evaluated test partition.

The findings are restricted to the dataset and experimental conditions examined. Independent external validation, repeated performance estimation, and further robustness analysis would be required before considering practical healthcare applications.

Overall, the assignment demonstrates a complete and reproducible supervised learning workflow and provides a structured comparison of conventional classifiers and advanced ensemble methods.

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